Hyperspectral Target Detection Based on Weighted Cauchy Distance Graph and Local Adaptive Collaborative Representation

Xiaobin Zhao, Wei Li, *Senior Member, IEEE*, Chunhui Zhao, and Ran Tao, *Senior Member, IEEE*

***Abstract*— Hyperspectral target detection in complex backgrounds is a challenging and important research topic in the remote sensing field. Traditional target detectors consider the background spectrum to obey a Gaussian distribution. However, this distribution may not meet the requirements in real hyperspectral images. In addition, the background and spatial information of most existing target detection algorithms are rarely fully utilized. Therefore, a new weighted Cauchy distance graph (WCDG) and local adaptive collaborative representation detection (CGCRD) is proposed. First, a WCDG similarity measure is designed. In order to adjust the effect of target pixels on the graph model, a weighted Cauchy distance Laplace matrix is constructed, and then the matrix is applied to the matched filter detector. Second, local adaptive collaborative representation strategy is developed. The penalty coefficient is weighted by the local spatial Euclidean distance combined with the Pearson correlation coefficient, and then the detection result is obtained based on the residual. Finally, aforementioned two strategies are fused to fully utilize the spatial and spectral information. A 176-band hyperspectral image (BIT-HSI-I) dataset is collected for the target detection task. The related algorithms are performed on the BIT-HSI-I dataset, and the detection results demonstrate that the proposed algorithm has better detection performance than other state-of-the-art algorithms.**

***Index Terms*— Euclidean distance, hyperspectral target detection, local adaptive collaborative representation, matched filter, Pearson correlation coefficient, weighted Cauchy distance.**

I. Introduction

HYPERSPECTRAL image covers both spatial information and spectral information of ground objects [1], [2]. Spectral information of tens or even hundreds and thousands of bands reflects the subtle differences between different materials, which lays the foundation for the rapid development of hyperspectral target detection and classification [3]. Target detection has great potential application value in many aspects, such as public safety, military reconnaissance, pollution monitoring, food hygiene, and so on. However, due to the influence of the sun's position, angle conditions, atmospheric conditions, and other factors, the background spectrum is complex and changeable [4]–[6]. The Gaussian distribution detection model may not be satisfactory. In addition, background statistics are easily contaminated by abnormal noise, which leads to high false alarm probability. Therefore, accurate hyperspectral target detection in complex background has become an arduous and challenging need in the field of remote sensing monitoring.

Based on whether the prior spectral information is adopted, hyperspectral target detection technologies are divided into anomaly detection and supervised target detection [7], [8]. Anomaly detection intends to locate targets that are very different from their surrounding background in spatial or spectrum under the unknown background and target spectral characteristics. Anomaly detection has the characteristics of blind detection, so it is unable to extract the clearly required target. Target detection uses prior spectral knowledge of the target or background to find pixels with higher similarity in the image [9]–[11]. This prior information can be obtained from laboratory measurements or extracted from images. Compared with anomaly detection, target detection algorithm has a certain pertinence and can extract specific desired targets [12]. According to the level of information utilization, hyperspectral target detection algorithms are roughly divided into spectral detection algorithms and spatial-spectral detection algorithms.

Spectral detection algorithms include probabilistic spatial models, subspace models, and nonlinear models. The probability model assumes that the spectrum satisfies a certain probability statistical model. Based on this model, statistical analysis is performed. Classical hypothesis testing methods include generalized likelihood ratio test (GLRT) [13]–[15]. The matching filter (MF) algorithm defined the target detection is also a hypothesis testing problem, and it has a good detection result when the background is simple [16]. When the detection image is large, the inversion of the covariance matrix of the algorithm will require a large amount of computation. Based on GLRT, the constant false alarm detection algorithm spectrum MF (SMF) algorithm was presented. Manolakis *et al.* [17] and Gao *et al.* [18] proposed an adaptive spectral MF (ASMF). Kraut *et al.* [19] developed the adaptive cosine estimation (ACE) by adding appropriate weighting factors to target noise and background noise

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covariance matrix. Farrand and Harsanyi [20] realized a constrained energy minimization (CEM) algorithm. It highlights the target and suppresses the background by using *a priori* spectral constraints to minimize the output energy [20]. Theiler and Foy [21] launched a covariance matrix estimation method for correcting the target signal. After removing outlier pixels, Basener [22] used the remaining background data to estimate the covariance matrix to obtain high parameter estimates. The subspace model increases the contrast between the target and the background to achieve target detection. The orthogonal subspace projection (OSP) algorithm achieved background suppression by projecting image elements into the background orthogonal subspace and improved the signal-to-noise ratio (SNR) of the signal [23]. Low probability detection (LPD) employed the eigenvectors corresponding to the correlation matrix to construct orthogonal subspaces. Among them, the spectral vector belonging to the background is suppressed, and the target spectral vector is enhanced after projection [24]. Wang *et al.* [25] obtained the unsupervised OSP (UOSP) by the method of convex geometry analysis. Estimating the virtual dimension accurately is the difficulty of the UOSP algorithm [26]–[28]. On the basis of the pixel linear decomposition analysis method, Heinz *et al.* [29] proposed the use of constrained least-squares for hyperspectral target detection. Kwon and Nasrabadi [30] extended the existing linear detection algorithm through the nuclear method to their corresponding nonlinear version. However, the Gaussian radial basis kernel function is commonly used in the current nuclear method, and there is a lack of rules for selecting the kernel function. Zhang *et al.* [31] applied the manifold learning method to hyperspectral target detection. They embed sparse transformation manifolds into frames to enable dimensionality reduction of spectral images. In the spatial after the dimensionality reduction, the nearest neighbor of the target feature vector was calculated to achieve the hyperspectral target detection [32].

With the advancement of hyperspectral sensors, hyperspectral imageries have not only higher spectral resolution but also higher spatial resolution. At present, there are few algorithms that simultaneously employ spatial spectral-domain information to detect targets. There is still a lot of room for improvement in target detection by making full use of spatial and spectral information and combining the advantages [33]. Kwon and Nasrabadi [34] proposed a kernel OSP (KOSP) method. In order to utilize the spatial neighborhood information, Li *et al.* [35] launched a hyperspectral anomaly detection method with combined sparse and collaborative representation (CSCR). The opinion of the algorithm is to use the target library and the background library to represent test pixels to achieve target detection. The tensor matching subspace detector treats hyperspectral imagery as a third-order tensor form to realize the joint utilization of multidimensional data information [36]. Yang *et al.* [37] proposed sparse-spatial CEM (SSCEM), which introduced a total variational regularization term in the CEM algorithm and forced the output to be sparse through the L1 norm. Although these algorithms use spatial information, the background model of the algorithm is fixed, which leads to poor robustness and a low detection probability. With the rise of deep learning, deep feature extraction has also been applied to hyperspectral target detection. The hyperspectral target detection network (HTD-NET) was developed [38]. The authors used the idea of pixel pairing to expand the amount of data and built a pixel similarity discrimination model. Shi *et al.* [39] designed a new deep-space spectral network (DSSN) framework for unsupervised hyperspectral target detection. This method uses classic detectors and edge-preserving filters to detect regions of interest. Cheng and Wang [40] presented a hyperspectral target detection decomposition model based on background dictionary learning (DM_BDL). They divided the image to be measured into three parts: target, background, and noise. The smoothness of the background pixel space and the sparseness of the target pixel were used, and the background dictionary learning method of locality-constrained linear coding was adopted. Zhang *et al.* [41] designed a subpixel target detection method based on supervised metric learning. The algorithm maximized the distance between the target and the background, and then used similarity propagation constraints to link target pixels and positive samples, background pixels, and negative samples to reduce false alarms. Among them, the manifold smoothness regularization method was adopted for positive samples.

In recent years, graph signal processing has been applied in many fields due to its ability to represent complex relationships between high-dimensional data. So far, graph-based methods have been applied to many fields, such as edge detection [42], gradient estimation [43], segmentation [44], [45], and so on. Graph theory has also recently been used to model relationships between samples [46], [47]. Considering that graph can models complex backgrounds and makes full use of background information, graph theory is considered for hyperspectral target detection. In order to overcome the shortcomings of existing algorithms such as poor background modeling, low spatial information utilization, and low detection probability, a weighted Cauchy distance graph (WCDG) and local adaptive collaborative representation detection (CGCRD) is proposed. The main contributions of the proposed CGCRD are as follows.

1) The hyperspectral target detector based on WCDG similarity measure is developed. The detector can not only avoid the problem of low detection rate caused by the hypothesis of the Gaussian distribution model for the hyperspectral background but also make full use of the relationship between the prior target pixels and testing pixels to reduce the effect of the target pixels on the background model. In addition, this method also avoids costly covariance matrix estimation and inversion.
2) A local adaptive collaborative representation algorithm for target detection is designed. The method adaptively utilizes spatial-spectral information to adjust the detection probability according to the Euclidean distance of the measured pixel and its correlation with the target spectrum.
3) The combination of spatial and spectral information improves the utilization of spatial-spectrum and improve

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the detection effect. The proposed CGCRD has superior detection performance on the constructed hyperspectral image (BIT-HSI-I) dataset. This dataset has been collected using the Dualix spectral imager GaiaSkymini2 on the campus of Beijing Institute of Technology (BIT). The details of this new dataset are presented in Section IV.

The remainder of this article is summarized as follows. Section II describes the related works. The proposed detection methodology is described in Section III which includes the WCDG similarity measure, adaptive collaborative representation, and spatial-spectrum fusion detection. In Section IV, evaluation methods are performed on the four hyperspectral images we acquired, which proves the superiority of the proposed algorithm. In Section V, the conclusion is given.

II. Related Works

A. Graph Signal Model

Graph signal models represent and handle high-dimensional data well. The most important thing for this model is to construct a weighting matrix [47], [48]. In an undirected weighted matrix, V and E denote the vertex sets and edge sets, respectively. The edge sets E can form a weight matrix $\boldsymbol{W}$. In an image, the vertexes V represent each pixel. The edges E is the weight parameter of the edge between V_a and V_b [49], [50]. It describes the mutual relationship between two pixels $\boldsymbol{x}_a$ and $\boldsymbol{x}_b$. When a and b are equal, the value of $\boldsymbol{W}_{ab}$ is 0.

After obtaining the weight matrix, $\boldsymbol{d}_q$ in the degree matrix represents the sum of the element values of the qth column of $\boldsymbol{W}$, namely $\boldsymbol{d}_q = \sum_{p=1}^{N} \boldsymbol{W}_{pq}$. The degree matrix is expressed as $\boldsymbol{D} = \text{diag}(\boldsymbol{d}_1, \ldots, \boldsymbol{d}_N)$. The diag represents a diagonal matrix, p, q refers to row and column, N is the number of image pixels. The graph Laplace matrix is

$$\boldsymbol{L} = \boldsymbol{D} - \boldsymbol{W}. \tag{1}$$

B. Collaborative Representation

The collaborative representation target detection (CRTD) algorithm is based on the spatial neighborhood of background pixels [51]. Different from the concept based on sparse representation, the collaborative representation means the current pixel by linearly combining the spatial neighborhood pixels around the pixel. Finally, the desired solutions are obtained by minimizing the L_2 norm.

In hyperspectral imagery, $\boldsymbol{x}$ represents the target pixel, N is the number of all pixels, and L is the number of bands. Here, s is the sum of the number of spatial pixels around $\boldsymbol{x}$. $\boldsymbol{X}_s \in \mathbb{R}^{L\times s}$ represents local pixels. When solving the weight vector of the collaborative representation, it is transformed into a minimization problem

$$\min \|\boldsymbol{x} - \boldsymbol{X}_s\boldsymbol{\omega}\|_2^2 \tag{2}$$

where $\boldsymbol{\omega}$ is the coefficient matrix. Adding the constraint restriction to $\boldsymbol{\omega}$, the objective function is rewritten as

$$\arg\min_{\boldsymbol{\omega}} \|\boldsymbol{x} - \boldsymbol{X}_s\boldsymbol{\omega}\|_2^2 + \lambda\|\boldsymbol{\omega}\|_2^2. \tag{3}$$

The parameter λ is the Lagrangian multiplier, so the objective function is replaced by the following formula:

$$\arg\min_{\boldsymbol{\omega}} \left[\boldsymbol{\omega}^T\left(\boldsymbol{X}_s^T\boldsymbol{X}_s + \lambda\boldsymbol{I}\right)\boldsymbol{\omega} - 2\boldsymbol{\omega}^T\boldsymbol{X}_s^T\boldsymbol{x}\right]. \tag{4}$$

The above formula is solved by

$$\boldsymbol{\omega} = \left(\boldsymbol{X}_s^T\boldsymbol{X}_s + \lambda\boldsymbol{I}\right)^{-1}\boldsymbol{X}_s^T\boldsymbol{x}. \tag{5}$$

The target is obtained by calculating the residual between the actual value and the predicted value. The residual is expressed as

$$\boldsymbol{r}_{\text{CRD}} = \|\boldsymbol{x} - \hat{\boldsymbol{x}}\|_2 = \|\boldsymbol{x} - \boldsymbol{X}_s\boldsymbol{\omega}\|_2 \tag{6}$$

where $\boldsymbol{x}$ and $\hat{\boldsymbol{x}}$ represent the actual and predicted values, respectively. $\boldsymbol{I}$ in (4) and (5) is the unit matrix.

III. Proposed Detection Methodology

The diagram of the proposed CGCRD is illustrated in Fig. 1. First, the Laplacian matrix based on the weighted Cauchy distance is calculated. The correlation between the prior target pixels and the background pixels is used to adjust the scale of target and background in the whole graph model. The symmetric normalized Laplacian matrix is employed to replace the covariance matrix of the matched filter to achieve target detection. Second, an algorithm based on local adaptive collaborative representation is designed. The penalty coefficient is weighted by the local spatial Euclidean distance combined with the Pearson correlation coefficient, and then the detection result is obtained using the residual. Finally, the two strategies are fused to get the final detection result.

A. WCDG Similarity Measure

Traditional detection methods are limited by the assumption of Gaussian model and rarely consider the influence of the target on the model, which will have a certain impact on the detection results. The Graph is often used to represent data, it describes the complex structure of data, especially for high-dimensional data have obvious advantages.

In the graph model, the measure of the edge between two vertices needs to be calculated. Considering that the Cauchy function has been shown to effectively capture the distance of a graph [52], the similarity or affinity of connected vertices can be better measured. Therefore, in order to better adapt to complex data and make full use of the correlation between bands, this thesis applies the Cauchy distance to the calculation of the graph edge weight matrix.

Considering hyperspectral image has N pixels and L bands. The hyperspectral imagery is expressed as an $L \times N$ matrix: $\boldsymbol{X} = (\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_i, \ldots, \boldsymbol{x}_N)$. The $\boldsymbol{X} \in \mathbb{R}^{L\times N}$, $\boldsymbol{x}_i \in \mathbb{R}^{L\times 1}, i = 1, 2, \ldots, N$. Assuming $\boldsymbol{t}$ ($\boldsymbol{t} \in \mathbb{R}^{L\times 1}$) is the target pixel, the Cauchy distance between different bands of a pixel is derived as

$$\boldsymbol{C}(i) = \frac{1}{1 + \frac{l_g - l_h}{\bar{l}}} \tag{7}$$

where l_g and l_h represent the reflected energy of band g and band h, respectively. The $\bar{l}$ represents the mean value of the band, it is expressed as $\bar{l} = (1/L)\sum_1^L l_k$.

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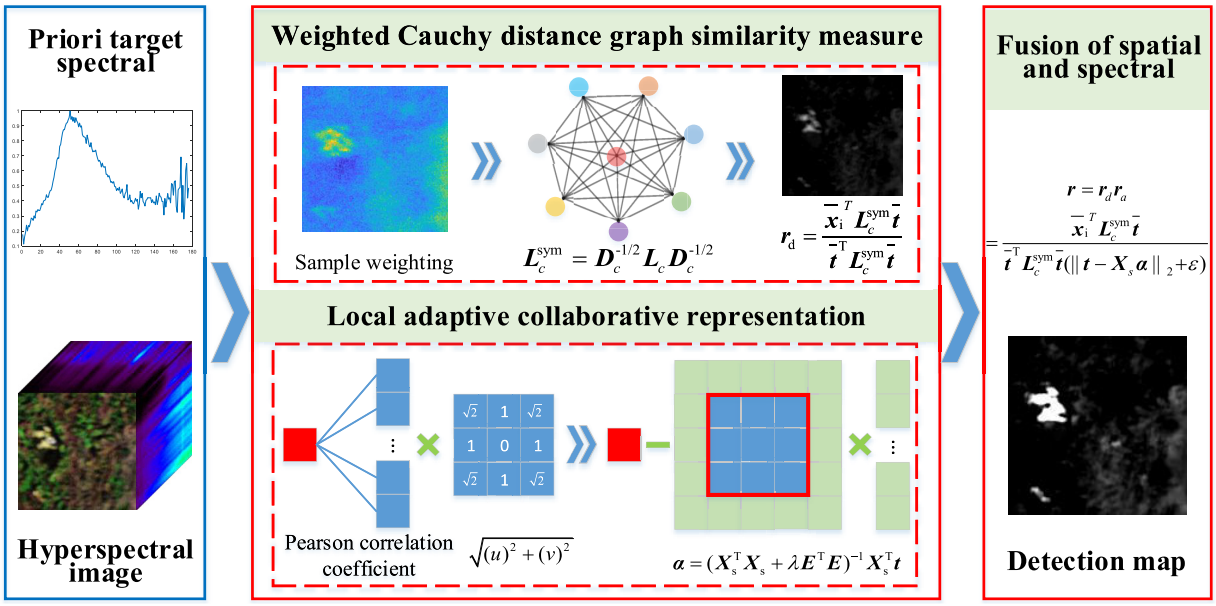


Fig. 1. Diagram of the proposed CGCRD. Including WCDG and local adaptive CGCRD.

The adjacency matrix $\boldsymbol{W}$ of the sample set is considered to be the weighted average of the adjacency matrices of all pixels $\boldsymbol{x}_i (i = 1, 2, \ldots, N)$. The subtle difference between the target spectrum under test and the prior target spectrum causes the filter results to be very different in the subsequent matched filter detection. Therefore, when calculating $\boldsymbol{W}$, the influence of the target signal of interest $\boldsymbol{t}$ in $\boldsymbol{W}$ is as small as possible, and the influence of signal with a large difference from $\boldsymbol{t}$ in $\boldsymbol{W}$ is as large as possible. For different samples, the target and background image should be weighted to determine their contribution to $\boldsymbol{W}$. The calculation formula of the correlation between the prior target spectrum and the background spectrum is as

$$o_i = \|\boldsymbol{t} - \boldsymbol{x}_i\|_2 \tag{8}$$

where o_i is the Euclidean distance between the target and background. The weighted Cauchy distance for a single background pixel is used to construct the adjacency matrix, and its formula is expressed as

$$\boldsymbol{W}_c(i) = \frac{1}{1 + \frac{o_i(l_g - l_h)}{\bar{l}}} \tag{9}$$

where g and h are the index of spectral bands, l_g and l_h represent the reflected energy of band g and band h, respectively. Fig. 2 illustrates the target and background spectral curves after correlation weighting. Fig. 2(a) illustrates the selected full band, Fig. 2(b) shows the band for 70 to 120. The blue dashed and solid lines represent the background spectral curves before and after weighting, respectively. The yellow curve is also a similar curve representing the target spectrum. It is found that the spectrum of target pixels is suppressed and the spectrum of background pixels is enhanced. Fig. 3 illustrates the maps of the sum of all bands. Fig. 3(a) and (b) represent the original data and data after weighting, respectively. In Fig. 3(b), the target is suppressed and the background is enhanced.

The adjacency matrix of the whole image is obtained by weighted averaging of the adjacency matrix of each pixel, and the formula is stated as

$$\boldsymbol{W}_c = \frac{\boldsymbol{W}_1 + \cdots + \boldsymbol{W}_N}{N} \tag{10}$$

where N represents the number of all target pixels in the image. After the weight matrix is obtained, the degree matrix $\boldsymbol{D}_c$ is presented as

$$\boldsymbol{D}_c = \operatorname{diag}(\boldsymbol{d}_1, \ldots, \boldsymbol{d}_q, \ldots, \boldsymbol{d}_L) \tag{11}$$

where $\boldsymbol{d}_q$ represents the sum of the element values of the qth column of the weighting matrix $\boldsymbol{W}_c$, L is the number of columns of the matrix $\boldsymbol{W}_c$. The graph Laplacian matrix is calculated as

$$\boldsymbol{L}_c = \boldsymbol{D}_c - \boldsymbol{W}_c. \tag{12}$$

The symmetric normalized Laplacian is denoted as

$$\boldsymbol{L}_c^{\text{sym}} = \boldsymbol{D}_c^{-1/2} \boldsymbol{L}_c \boldsymbol{D}_c^{-1/2}. \tag{13}$$

Generally, a random vector $\boldsymbol{x} = [x_1, \ldots, x_L]$ satisfies a Gaussian Markov random field (MRF). The mean of the probability model of the graph structure $G = (V, E)$ is μ, and the precision matrix is $\boldsymbol{Q} > 0$. The density function satisfies

$$p(\boldsymbol{x}) = (2\pi)^{-\frac{N}{2}} |\boldsymbol{Q}|^{\frac{1}{2}} \exp\left(-\frac{1}{2}(\boldsymbol{x} - \mu)^T \boldsymbol{Q}(\boldsymbol{x} - \mu)\right)$$
$$\mathrm{Q}_{ij} \neq 0 \Leftrightarrow \{i, j\} \in E \quad i \neq j \tag{14}$$

where $\boldsymbol{Q}$ ($\boldsymbol{Q} = \delta \boldsymbol{L}$, δ is a scale factor) matrix represents the inverse of the multivariate covariance matrix [53].

Because the Laplace matrix is equivalent to the inverse of the multivariate covariance matrix. Applying the improved Laplacian matrix $\boldsymbol{L}_c$ to the matched filter detection makes

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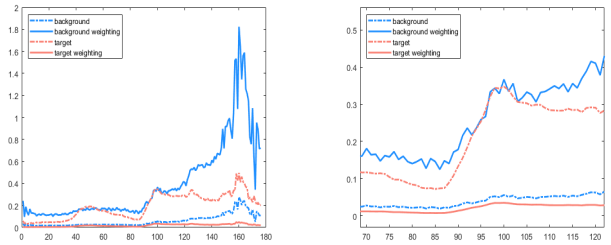


Fig. 2. Target and background spectral curves after correlation weighting. (a) Full band. (b) Local band.

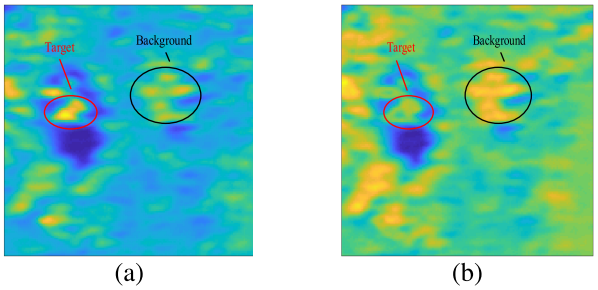


Fig. 3. Maps of the sum of all bands of the hyperspectral image. (a) Original data. (b) Data after weighting.

it more robust and reduces the computational complexity of its covariance matrix inversion. The detection output in the spectral domain is described as

$$r_d = \frac{\bar{\boldsymbol{x}}_i^T \boldsymbol{L}_c^{\text{sym}} \bar{\boldsymbol{t}}}{\bar{\boldsymbol{t}}^T \boldsymbol{L}_c^{\text{sym}} \bar{\boldsymbol{t}}} \tag{15}$$

where r_d is the detection result in the spectral domain. Here, $\bar{\boldsymbol{x}}_i$ indicates the data to be measured after removing the mean, $\bar{\boldsymbol{t}}$ represents the target of removing the mean.

B. Local Adaptive Collaborative Representation

The collaborative representation does not need to consider the background model and can make full use of the background information, so it is applied to the detection of hyperspectral targets. To further adjust the detection results, the Euclidean distance of the measured pixels and their correlation with the target spectrum is employed. This strategy uses a single-window collaborative representation to detect targets to reduce detection time. The target pixel $\boldsymbol{t}$ with the size of $L \times 1$ is approximated as

$$\hat{\boldsymbol{t}} = a_1 \boldsymbol{x}_1 + \alpha_2 \boldsymbol{x}_2 + \cdots + \alpha_s \boldsymbol{x}_s = \boldsymbol{X}_s \boldsymbol{\alpha} \tag{16}$$

where $\hat{\boldsymbol{t}}$ represents the estimated value of the pixel vector, $\boldsymbol{X}_s$ are the pixels in the window and is expressed as $\boldsymbol{X}_s = [\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_s]$. Here, $\boldsymbol{\alpha}$ is the coefficient matrix, and it is expressed as $\boldsymbol{\alpha} = [\alpha_1, \alpha_2, \ldots, \alpha_s]^T$, where s represents the number of background pixels within dual-window (w) and is calculated by $s = w^2$.

Note that, $\boldsymbol{X}_s$ is adaptively adjusted for each test pixel. The $\boldsymbol{\alpha}$ is obtained by the function

$$\arg\min_{\boldsymbol{\alpha}} \|\boldsymbol{t} - \boldsymbol{X}_s \boldsymbol{\alpha}\|_2^2 + \lambda \|\boldsymbol{\alpha}\|_2^2 \tag{17}$$

where $\boldsymbol{X}_s$ represents the purified background, λ is Lagrange multiplier. The weighted coefficient vector $\boldsymbol{\alpha}$ is calculated as

$$\hat{\boldsymbol{\alpha}} = \left(\boldsymbol{X}_s^T \boldsymbol{X}_s + \lambda \boldsymbol{I}\right)^{-1} \boldsymbol{X}_s^T \boldsymbol{t} \tag{18}$$

where $\boldsymbol{I}$ is the unit matrix. The coefficients are small for those pixels being tested that have large differences from the prior pixels, so the penalty parameter must be increased for larger coefficients. Therefore, Euclidean distance combined with Pearson correlation coefficient is utilized to weight the penalty

$$\boldsymbol{E} = \begin{bmatrix} e_1 p_1 & \cdots & 0 \\ & \ddots & \\ \vdots & e_k p_k & \vdots \\ & & \ddots \\ 0 & \cdots & e_s p_s \end{bmatrix}. \tag{19}$$

The position coordinates of $\boldsymbol{x}_i$ is represented as $(0, 0)$. The coordinate of the kth pixel $\boldsymbol{x}_k$ $(k = 1, 2, \ldots, s)$ in the local window s is (u, v). Then, the Euclidean distance between $\boldsymbol{x}_k$ and $\boldsymbol{x}_i$ is expressed by $e_k(p, q) = (u^2 + v^2)^{1/2}$. The p_k is the derivative of $\rho_{\boldsymbol{t}\boldsymbol{x}_k}$, and it is denoted as

$$p_k = \frac{1}{\rho_{\boldsymbol{t}\boldsymbol{x}_k}}. \tag{20}$$

The correlation coefficient $\rho_{\boldsymbol{t}\boldsymbol{x}_k}$ between the kth pixel and the prior target pixel is stated as

$$\rho_{\boldsymbol{t}\boldsymbol{x}_k} = \frac{\sum \boldsymbol{t}^T \boldsymbol{x}_k - \frac{\sum \boldsymbol{t} \sum \boldsymbol{x}_k}{L}}{\sqrt{\left(\sum \boldsymbol{x}_k^T \boldsymbol{x}_k - \frac{\left(\sum \boldsymbol{x}_k\right)^2}{L}\right)\left(\sum \boldsymbol{t}^T \boldsymbol{t} - \frac{\left(\sum \boldsymbol{t}\right)^2}{L}\right)}}. \tag{21}$$

The objective function after adding Euclidean distance and Pearson correlation coefficient regularization is rewritten as

$$\arg\min_{\boldsymbol{a}} \|\boldsymbol{t} - \boldsymbol{X}_s \boldsymbol{\alpha}\|_2^2 + \lambda \|\boldsymbol{E}\boldsymbol{\alpha}\|_2^2. \tag{22}$$

By using the Lagrange multiplier method, a final solution is obtained by

$$\hat{\boldsymbol{\alpha}} = \left(\boldsymbol{X}_s^T \boldsymbol{X}_s + \lambda \boldsymbol{E}^T \boldsymbol{E}\right)^{-1} \boldsymbol{X}_s^T \boldsymbol{t}. \tag{23}$$

In target detection, the smaller the residual value, the closer to the target. The detection output is

$$r_a = \frac{1}{\|\boldsymbol{t} - \boldsymbol{X}_s \hat{\boldsymbol{\alpha}}\|_2 + \varepsilon} \tag{24}$$

where ε is a constant. It avoids having no solution when the denominator is 0 (residual is 0).

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TABLE I
HYPERSPECTRAL TARGET DETECTION EXPERIMENTAL DATASETS COLLECTED BY THE BIT GROUP

Hyperspectral image	Collection date	Spectral resolution	The size of spatial	Sensor	Background introduction
Imagery 1	23/10/2021	3.5nm ± 0.5nm	150×150	GaiaSkymini2	Wet grassland
Imagery 2	31/10/2021	3.5nm ± 0.5nm	150×150	GaiaSkymini2	Dry grassland
Imagery 3	08/11/2021	3.5nm ± 0.5nm	150×150	GaiaSkymini2	Shaded grassland
Imagery 4	12/10/2021	3.5nm ± 0.5nm	150×150	GaiaSkymini2	Leaf ground

C. Spatial Spectral Fusion Detection

In order to use spectral and spatial information at the same time, this thesis adopts the method of fusion of spectral-domain information and spatial-domain information to achieve target detection. The final output is described as

$$r = r_d r_a = \frac{\bar{\boldsymbol{x}}_i^T \boldsymbol{L}_c^{\text{sym}} \bar{\boldsymbol{t}}}{\bar{\boldsymbol{t}}^T \boldsymbol{L}_c^{\text{sym}} \bar{\boldsymbol{t}} (\|\boldsymbol{t} - \boldsymbol{X}_s \hat{\boldsymbol{\alpha}}\|_2 + \varepsilon)}. \tag{25}$$

If r exceeds the preset threshold, it is judged as the desired target pixel. In addition to that, r is determined to be the background pixel. The framework of the proposed CGCRD is summarized in Algorithm 1.

Algorithm 1 Main Steps of the Proposed CGCRD Algorithm

Input: $\boldsymbol{X}$: The hyperspectral imagery; $\boldsymbol{t}$: prior target pixels; λ: the Lagrange multiplier; w_{in}: the size of window.
Output: The detection result $\boldsymbol{r} = [r_1, \ldots, r_N]$.

1: **for** $i = 1 \ldots N$ **do**
2: The similar between the target spectrum and background spectrum is calculated by the formula (8);
3: The adjacency matrix is calculated by formula (10);
4: The Laplace matrix is calculated by formula (13);
5: The detection output in spectral domain is obtained by formula (15);
6: The Euclidean distance combined with Pearson correlation coefficient between prior target and background is calculated by formula (19);
7: The coefficient of collaborative representation is obtained by formula (23);
8: The detection output in spatial domain is obtained by formula (24);
9: The result of spatial-spectral detection is obtained by formula (25);
10: **end for**

IV. EXPERIMENTS AND ANALYSIS

A. Hyperspectral Datasets

A new BIT-HSI-I dataset has been collected to perform the experiments. The prior target spectrum is obtained from the close-range camouflage car spectrum of the corresponding scene collected in advance. Here, a center pixel spectrum is chosen as the prior. The dataset was acquired using the Dualix spectral imager GaiaSkymini2 on the campus of the BIT. The spectral range is 400–1000 nm, and the spectral resolution is 3.5 nm. The spatial resolution at a distance of 500 m is 0.14 m. The collected target is a camouflage vehicle model. The backgrounds are wet grassland, dry grassland, shaded grassland, and leaf ground, respectively. The detailed descriptions of the dataset are presented in Table I. Figs. 4–7 display the pseudo-color images and the ground truth maps.



(a) (b)

Fig. 4. Wet grassland image. (a) Pseudo-color image. (b) Ground-truth map.



(a) (b)

Fig. 5. Dry grassland image. (a) Pseudo-color image. (b) Ground-truth map.

B. Parameters Analysis

We validate the effect of the proposed CGCRD using four different scenarios. To demonstrate the superiority of CGCRD, it is contrasted with existing seven superior methods. Seven algorithms compared are ACE [19], CEM [20], MF [16], ASMF [17], CSCR [35], SSCEM [37], and DM_BDL [40]. Next, the parameters and experimental results of the proposed algorithm are analyzed, respectively.

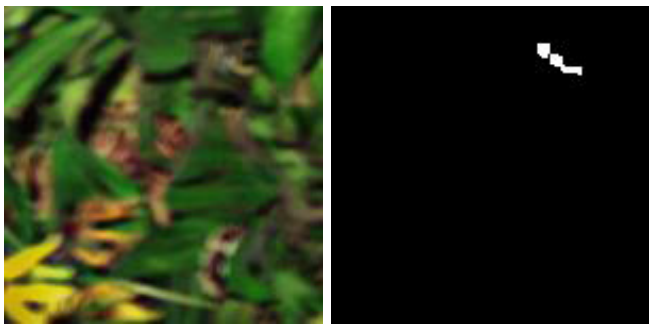
By varying the parameter λ and the window size w_{in}, Fig. 8 illustrates the area under the curve (AUC) of the proposed CGCRD. Fig. 8(a) displays the AUC values under different λ and w_{in} for the wet grassland image. For example, when λ and w_{in} are selected as 10^{-4} and 15, respectively, the proposed CGCRD has the best detection performance with an AUC value of 0.9974. When w_{in} is selected as 5, the AUC value becomes smaller and drops to 0.9950. Fig. 8(b) displays the AUC values under different λ and w_{in} for the dry

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(a) (b)

Fig. 6. Shaded grassland image. (a) Pseudo-color image. (b) Ground-truth map.



(a) (b)

Fig. 7. Leaf ground image. (a) Pseudo-color image. (b) Ground-truth map.

grassland image. For example, when λ and w_{in} are selected as 10^{-1} and 3, respectively, the proposed CGCRD has the best detection performance with an AUC value of 0.9992. When w_{in} is selected as 9, the AUC value becomes smaller and drops to 0.9990. Fig. 8(c) displays the AUC values under different λ and w_{in} for the shaded grassland image. For example, when λ and w_{in} are selected as 10^{-5} and 9, respectively, the proposed CGCRD has the best detection performance with an AUC value of 0.9990. When w_{in} is selected as 3, the AUC value becomes smaller and drops to 0.9882. Fig. 8(d) displays the AUC values under different λ and w_{in} for the leaf ground image. For example, when λ and w_{in} are selected as 10^{-5} and 11, respectively, the proposed CGCRD has the best detection performance with an AUC value of 0.9941. When w_{in} is selected as 3, the AUC value becomes smaller and drops to 0.9714. Through the above analysis, it is found that the value of λ has little change in detection performance. The value of λ is set to 10^{-5}. The value of w_{in} is similar to the number of target pixels in the image. Its value is chosen based on the number of these pixels.

Fig. 9 displays the receiver operating characteristic curve (ROC) curves of the proposed Cauchy similarity measure, the weighted Cauchy similarity measure, and the spatial-spectral fusion method, respectively. All four datasets demonstrate a progressive improvement in the detection performance of the proposed method.

C. Detection Performance

To quantitatively and qualitatively compare the proposed method, detection map, target background separation box, and

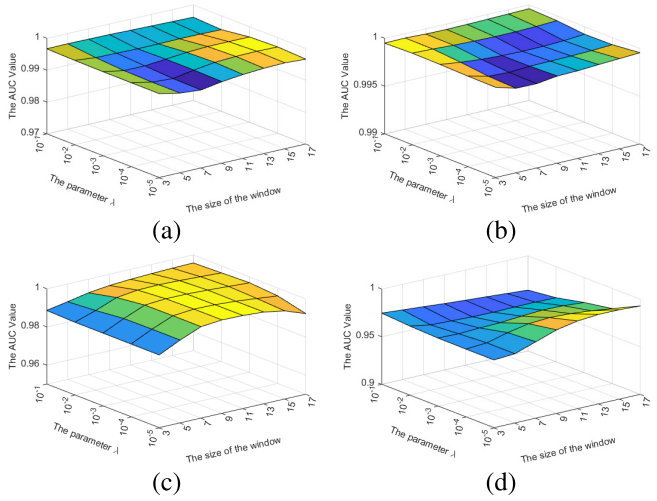


Fig. 8. AUC values for different λ and window size under different datasets. (a) Wet grassland. (b) Dry grassland. (c) Shaded grassland. (d) Leaf ground.

ROC are introduced. The detection map is a gray map. The brighter the pixel, the closer to the target.

The normalized values corresponding to the target and background locations in the detection probability are grouped into two groups. Dividable boxes represent statistical intervals of target and background normalized values as different colored boxes [4]. The position of the horizontal line in the middle of each box is the median, the upper and lower edges are the 75th and 25th of the maximum value of the data. The extremum of the detection result is represented by the "whisker" [54].

The abscissa and ordinate of the ROC curve are plotted with the probability of false detection (PF) and the probability of detection (PD), respectively. The closer the ROC is to the northwest corner of the figure, the higher the PD and the lower the PF, indicating that algorithm involved performance is more excellent. The AUC refers to the area enclosed by the ROC and the coordinate axis. It is expressed as

$$\text{AUC} = \frac{1}{2} \times \sum_{i}^{w} (\text{PF}_i - \text{PF}_{i-1}) \times (\text{PD}_i + \text{PD}_{i-1}) \tag{26}$$

where PD is detection probability, PF is false alarm probability, w is the number of different thresholds. The PD denotes the percentage of correctly positioned target pixels to the whole number of target pixels actually possessed. The PF is the percentage of nontarget pixels that are considered target pixels to the whole number of pixels. Under a certain PF, the higher the PD, the better the function of the algorithm used [55], [56]. The PD and PF are calculated by

$$\text{PD} = \frac{N_d}{N_t} \tag{27}$$

$$\text{PF} = \frac{N_f}{N} \tag{28}$$

where N_d, N_t, N_f, and N indicate the correct number of pixels are detected, the real target pixels number of image, wrongly detected pixels, and all pixels of the image, respectively.

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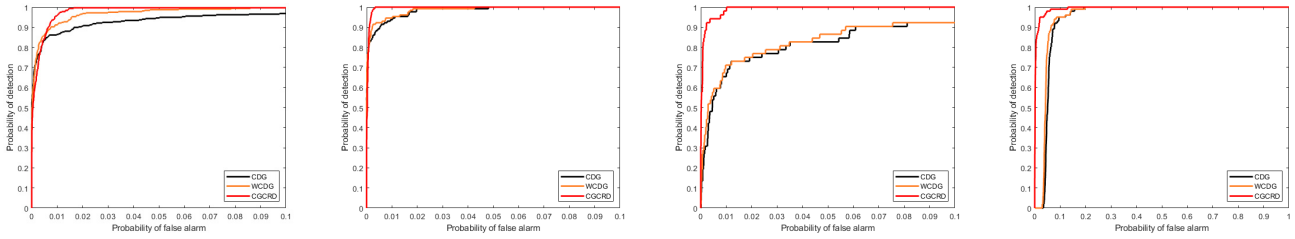


Fig. 9. ROC curves of the proposed CDG, WCDG and CGCRD in different scenes. (a) Wet grassland. (b) Dry grassland. (c) Shaded grassland. (d) Leaf ground.

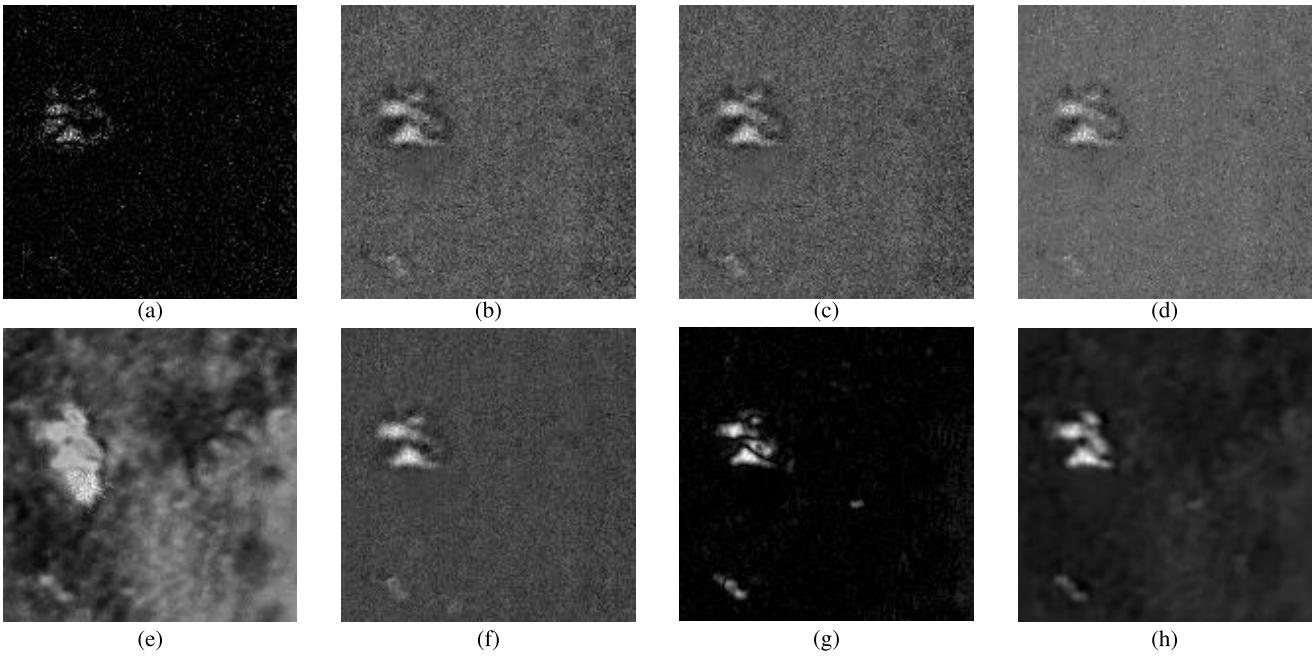


Fig. 10. Detection results of eight methods in real wet grassland. (a) ACE. (b) CEM. (c) MF. (d) ASMF. (e) CSCR. (f) SSCEM. (g) DM_BDL. (h) CGCRD.

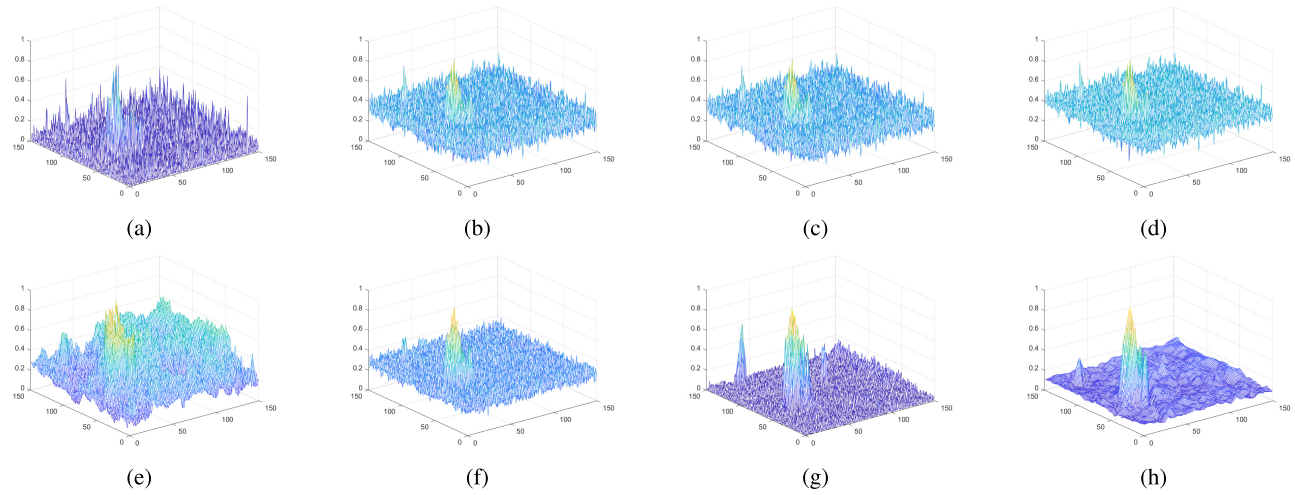


Fig. 11. 3-D detection results of eight methods in real wet grassland. (a) ACE. (b) CEM. (c) MF. (d) ASMF. (e) CSCR. (f) SSCEM. (g) DM_BDL. (h) CGCRD.

Figs. 10–17 illustrate the results of 2D and 3D visual detection by different algorithms in four scenarios. The proposed algorithm is able to identify more target pixels in the datasets of wet grass, dry grass, shaded

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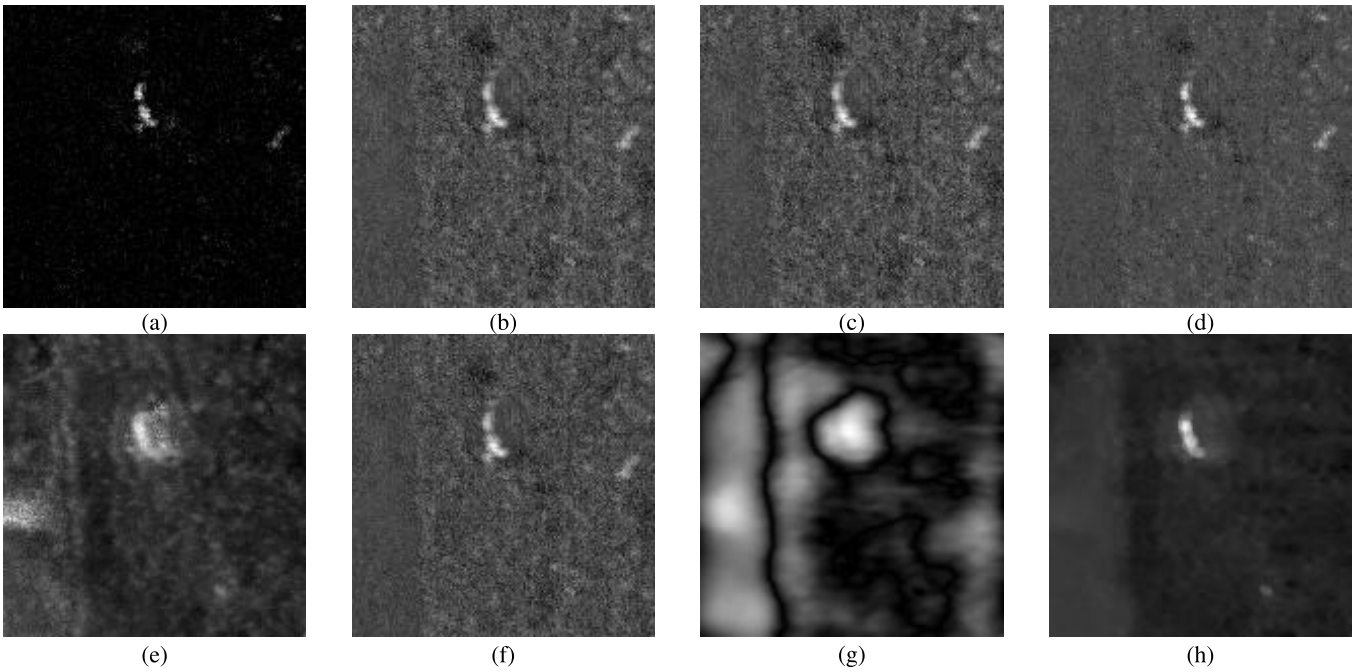


Fig. 12. Detection results of eight methods in real dry grassland. (a) ACE. (b) CEM. (c) MF. (d) ASMF. (e) CSCR. (f) SSCEM. (g) DM_BDL. (h) CGCRD.

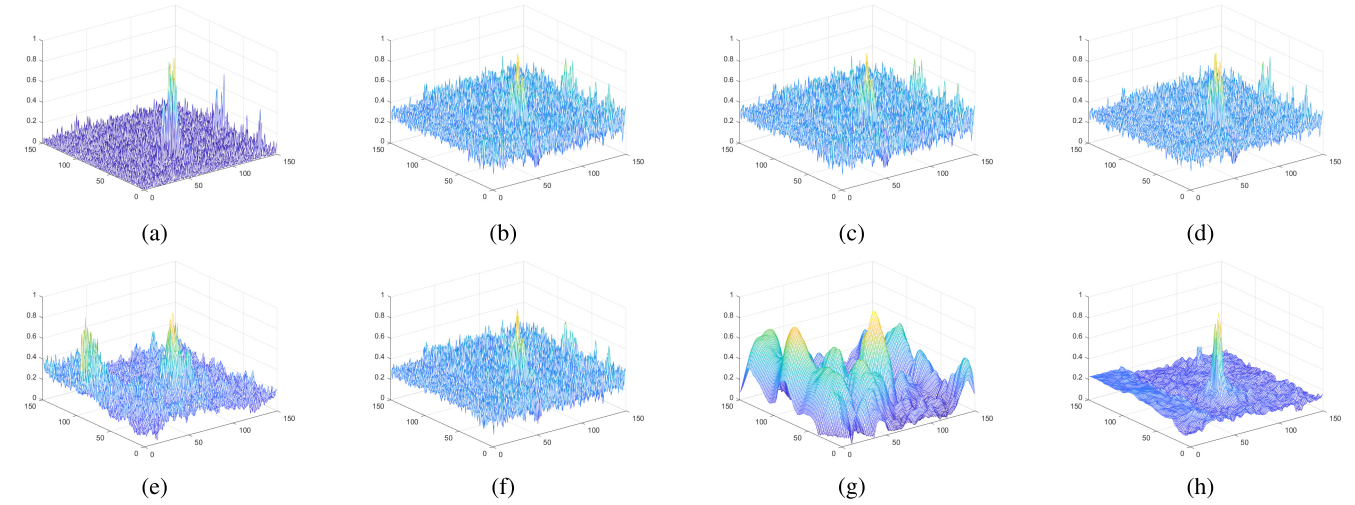


Fig. 13. 3-D detection results of eight methods in real dry grassland. (a) ACE. (b) CEM. (c) MF. (d) ASMF. (e) CSCR. (f) SSCEM. (g) DM_BDL. (h) CGCRD.

grass, and leafy fields. Other comparison algorithms have poor detection performance. In Figs. 10 and 11(a)–(d), ACE, CEM, MF, ASMF detect fewer correct pixels. In Figs. 10 and 11(e)–(g), CSCR, SSCEM, DM_BDL detect more target pixels, but there are also many wrong target pixels. In Figs. 12 and 13(a), although the ACE algorithm has fewer false alarms, it detects fewer correct pixels. In Figs. 12 and 13(b)–(g), CEM, MF, ASMF, CSCR, SSCEM, and DM_BDL detect more wrong target pixels, and DM_BDL detects the most wrong pixels. In Figs. 14 and 15(a)–(f), ACE, CEM, MF, ASMF, CSCR, SSCEM detect fewer wrong target pixels and fewer correct target pixels. In Figs. 14 and 15(g), DM_BDL detects some correct pixels, and detect many wrong pixels. In Figs. 16 and 17(a), (d), and (g), ACE, ASMF, DM_BDL hardly detect the target pixels. In Figs. 16 and 17(b), (c), (e), and (f), CEM, MF, CSCR, SSCEM detect fewer wrong target pixels and fewer correct target pixels. Therefore, the proposed CGCRD always has the superior detection performance, which increases the separation of target and background.

In the separation box, the red squares represent the target probability area, and the green squares represent the background probability area. The space between the red squares and the green squares represents the separation from the target to the background. In Fig. 18(a)–(d), the spacing between red and green boxes of CGCRD is larger

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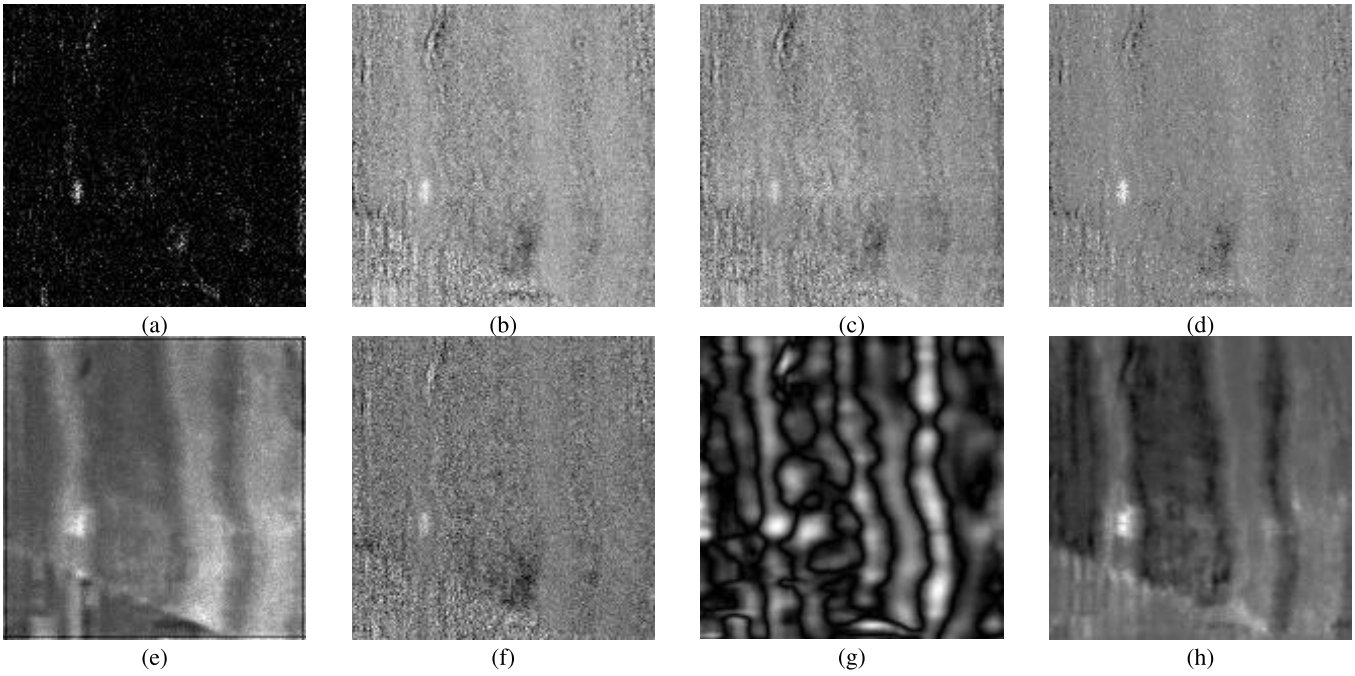


Fig. 14. Detection results of eight methods in real shaded grassland. (a) ACE. (b) CEM. (c) MF. (d) ASMF. (e) CSCR. (f) SSCEM. (g) DM_BDL. (h) CGCRD.

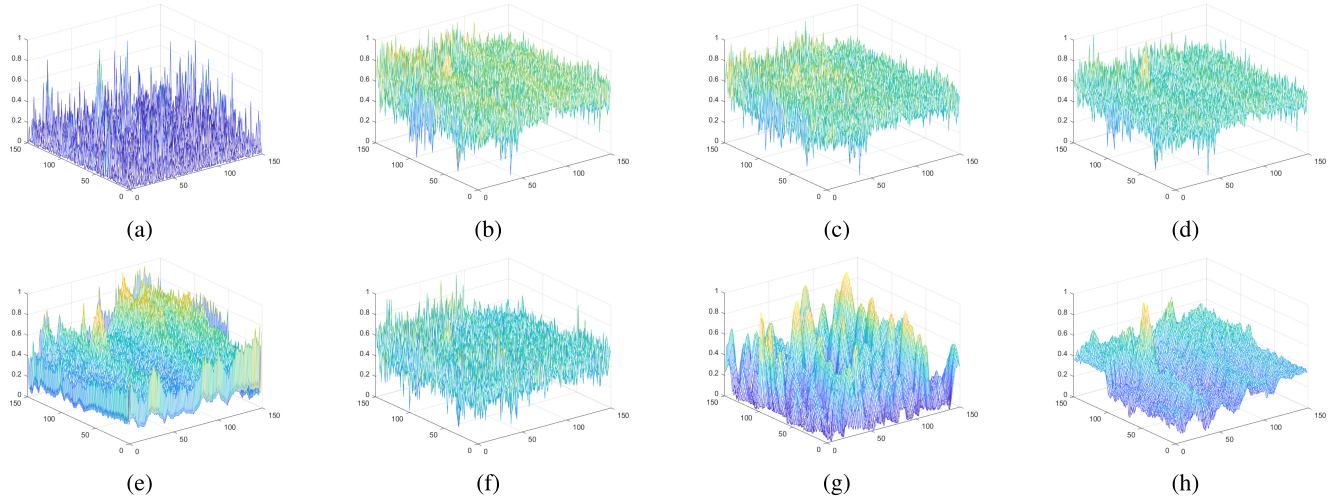


Fig. 15. 3-D detection results of eight methods in real shaded grassland. (a) ACE. (b) CEM. (c) MF. (d) ASMF. (e) CSCR. (f) SSCEM. (g) DM_BDL. (h) CGCRD.

than that of other algorithms. This shows that the proposed CGCRD easily identifies the target from the background image.

The coordinates on the ROC curve indicate PD and PF at different thresholds. In Fig. 19(a), which is the wet grass image, when PF is are all 0.1, the PD corresponding to CGCRD is about 1.00. The PD of other methods (ACE, CEM, MF, ASMF, CSCR, SSCEM, DM_BDL) are 0.5394, 0.6909, 0.6940, 0.6341, 0.9905, 0.7319, and 0.9558, respectively. From Fig. 19(b)–(d), when PF is 0.01, 0.01 and 0.1, the PD of CGCRD are 1.0, 0.9808, and 1.0, respectively. The maximum PD of other mentioned algorithms are 0.9147, 0.6731, and 0.9901, respectively. These results demonstrate the effectiveness of the proposed CGCRD.

Table II lists the AUC values of the eight comparison algorithms. The AUC values of CGCRD are 0.9974, 0.9992, 0.9990, and 0.9941 for these four scenarios, respectively. The optimal AUC values of the remaining algorithms in the four different scenarios are 0.9846, 0.9968, 0.9764, and 0.9861, respectively. These values are poor relative to the AUC values of CGCRD. The running times of the eight detection algorithms are shown in Table III. In four different scenarios, the time used by CGCRD is 29.9725, 5.1503, 25.5902, and 67.1497 s, respectively. The time consumption of CGCRD

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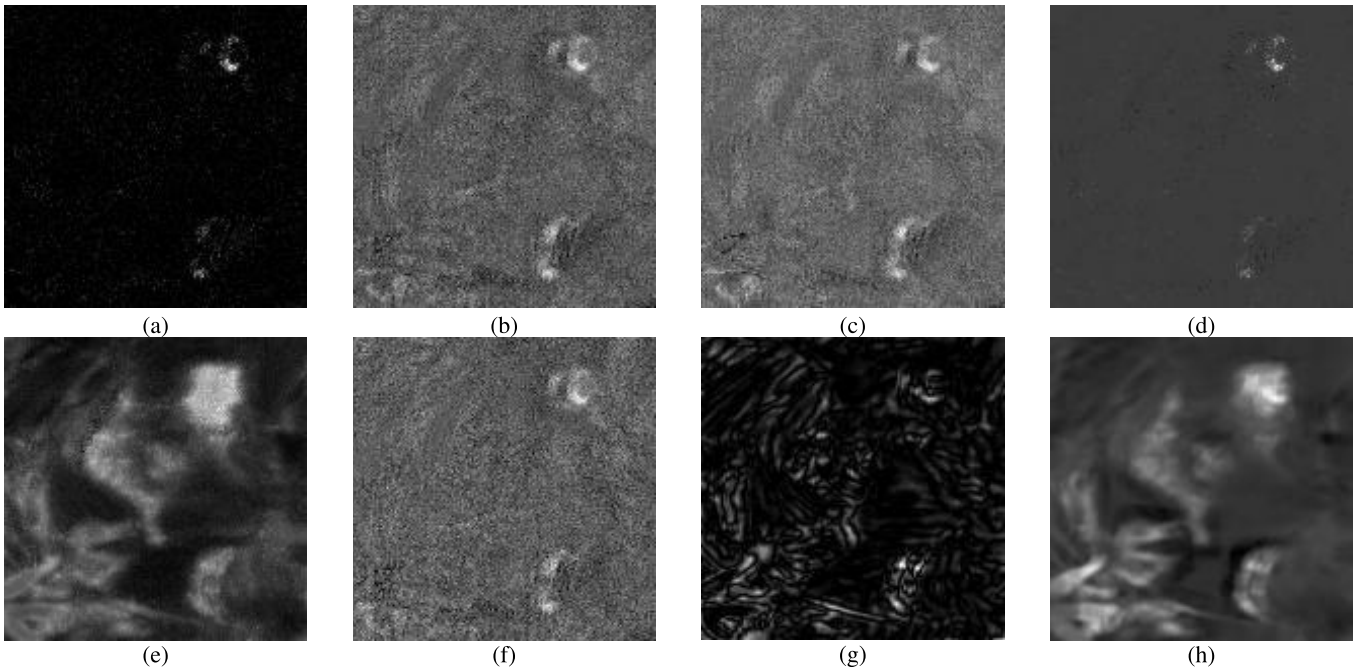


Fig. 16. Detection results of eight methods in real leaf ground. (a) ACE. (b) CEM. (c) MF. (d) ASMF. (e) CSCR. (f) SSCEM. (g) DM_BDL. (h) CGCRD.

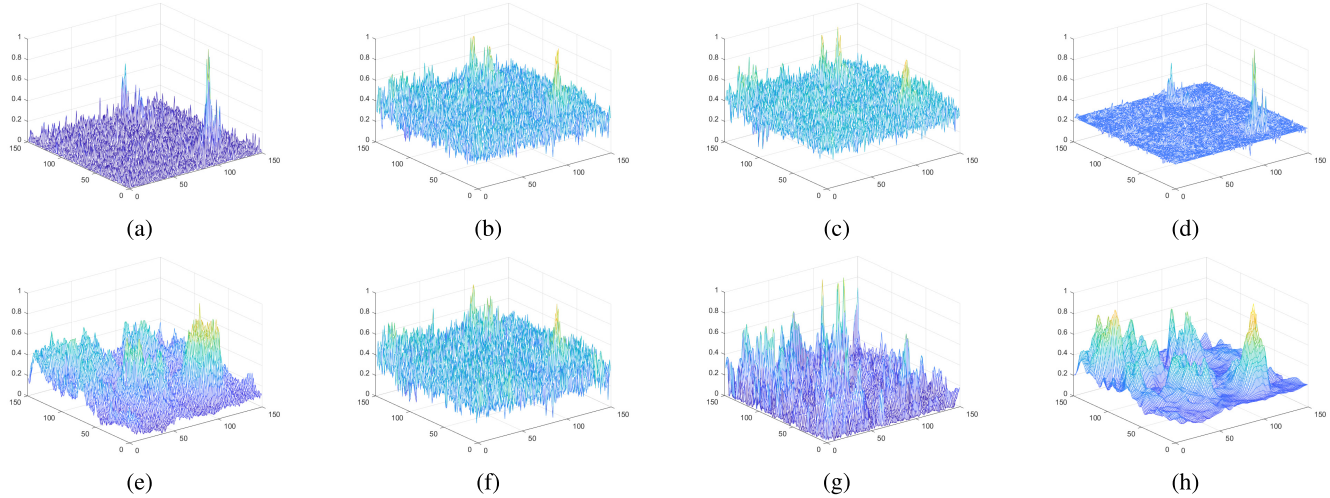


Fig. 17. 3-D detection results of eight methods in real leaf ground. (a) ACE. (b) CEM. (c) MF. (d) ASMF. (e) CSCR. (f) SSCEM. (g) DM_BDL. (h) CGCRD.

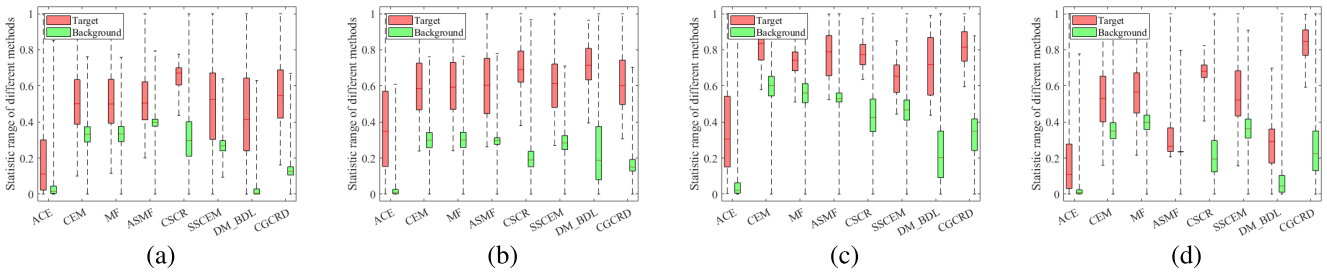


Fig. 18. Statistical separability analysis with different detectors using four experimental datasets. (a) Wet grassland. (b) Dry grassland. (c) Shaded grassland. (d) Leaf ground.

is mainly used for the local window calculation method. In addition, the time consumption of SSCEM is the largest. The computational efficiency of our proposed algorithm is also good. All tests algorithms are performed with MATLAB on an Intel Core i7-8700 h CPU computer with 8 Gb of computer memory.

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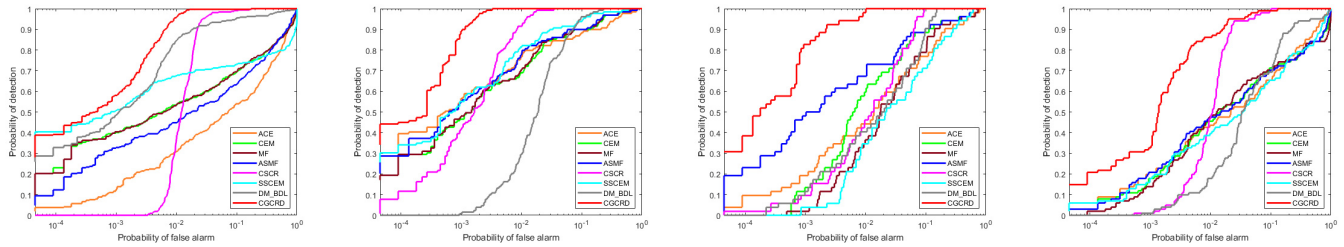


Fig. 19. ROC curve with different detectors using four experimental datasets. (a) Wet grassland. (b) Dry grassland. (c) Shaded grassland. (d) Leaf ground.

TABLE II
AUC VALUES OF DIFFERENT DETECTORS ON FOUR DATASETS

Image	ACE	CEM	MF	ASMF	CSCR	SSCEM	DM_BDL	CGCRD
Wet grassland	0.7634	0.8296	0.8314	0.8217	0.9846	0.7976	0.9739	0.9974
Dry grassland	0.9489	0.9618	0.9634	0.9650	0.9968	0.9798	0.9705	0.9992
Shaded grassland	0.9164	0.9510	0.9280	0.9559	0.9764	0.9071	0.9620	0.9990
Leaf ground	0.8487	0.8137	0.8053	0.8162	0.9861	0.8253	0.8966	0.9941

TABLE III
EXECUTION TIME OF DIFFERENT DETECTORS ON FOUR DATASETS (UNIT: SECONDS)

Image	ACE	CEM	MF	ASMF	CSCR	SSCEM	DM_BDL	CGCRD
Wet grassland	0.2031	0.0949	0.1302	1.3822	4.5834	231.0282	4.5004	29.9725
Dry grassland	0.1966	0.0839	0.1302	1.1616	4.4626	222.1655	7.0287	5.1503
Shaded grassland	0.1882	0.0825	0.1263	1.1656	8.0926	228.3280	5.4691	25.5902
Leaf ground	0.2140	0.0806	0.1330	1.2282	4.4288	203.2471	4.5716	67.1497

V. CONCLUSION

In this article, an effective hyperspectral target detection algorithm was proposed based on weighted Cauchy distance graph (CDG) and locally adaptive collaborative representation. The algorithm has employed the Laplacian matrix of weighted Cauchy distance to estimate background statistics, which improves the adaptability of the algorithm to complex datasets and reduces the calculation amount of covariance matrix estimation and inversion. Meanwhile, in order to fully utilize the local spatial information, an adaptive collaborative representation method was developed. It adopts the Euclidean distance combined with the Pearson correlation coefficient to adaptively penalize the collaborative representation coefficient, which can adjust the performance of the target detection. These experimental results show that the proposed CGCRD has better detection performance compared to other detection algorithms on four real hyperspectral imageries, and the detection performance of the proposed CGCRD is up to 30% higher than the comparison method using the collected BIT-HSI-I data.

REFERENCES

[1] R. O. Green *et al.*, "Imaging spectroscopy and the airborne visible/infrared imaging spectrometer," *Remote Sens. Environ.*, vol. 65, no. 3, pp. 227–248, Sep. 1998.

[2] R. Tao, X. Zhao, W. Li, H. Li, and Q. Du, "Hyperspectral anomaly detection by fractional Fourier entropy," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 12, pp. 4920–4929, Dec. 2019.

[3] N. M. Nasrabadi, "Hyperspectral target detection: An overview of current and future challenges," *IEEE Signal Process. Mag.*, vol. 31, no. 1, pp. 34–44, Jan. 2014.

[4] Z. Hou, W. Li, L. Li, R. Tao, and Q. Du, "Hyperspectral change detection based on multiple morphological profiles," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–12, 2021.

[5] A. F. H. Goetz, G. Vane, J. E. Solomon, and B. N. Rock, "Imaging spectrometry for earth remote sensing," *Science*, vol. 228, no. 4704, pp. 1147–1153, 1985.

[6] J. M. Bioucas-Dias, A. Plaza, G. Camps-Valls, P. Scheunders, N. M. Nasrabadi, and J. Chanussot, "Hyperspectral remote sensing data analysis and future challenges," *IEEE Geosci. Remote Sens. Mag.*, vol. 1, no. 2, pp. 6–36, Jun. 2013.

[7] D. Landgrebe, "Hyperspectral image data analysis," *IEEE Signal Process. Mag.*, vol. 19, no. 1, pp. 17–28, Jan. 2002.

[8] Y. Dong, B. Du, L. Zhang, and X. Hu, "Hyperspectral target detection via adaptive information—Theoretic metric learning with local constraints," *Remote Sens.*, vol. 10, no. 9, p. 1415, Sep. 2018.

[9] W. Xie, X. Zhang, Y. Li, K. Wang, and Q. Du, "Background learning based on target suppression constraint for hyperspectral target detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 5887–5897, 2020.

[10] L. Li, W. Li, Q. Du, and R. Tao, "Low-rank and sparse decomposition with mixture of Gaussian for hyperspectral anomaly detection," *IEEE Trans. Cybern.*, vol. 51, no. 9, pp. 4363–4372, Sep. 2021.

[11] K. Tan, Z. Hou, F. Wu, Q. Du, and Y. Chen, "Anomaly detection for hyperspectral imagery based on the regularized subspace method and collaborative representation," *Remote Sens.*, vol. 11, no. 11, p. 1318, Jun. 2019.

[12] X. Zhao, W. Li, M. Zhang, R. Tao, and P. Ma, "Adaptive iterated shrinkage thresholding-based L_p-norm sparse representation for hyperspectral imagery target detection," *Remote Sens.*, vol. 12, no. 23, p. 3991, Dec. 2020.

[13] S. Kraut, L. L. Scharf, and L. T. McWhorter, "Adaptive subspace detectors," *IEEE Trans. Signal Process.*, vol. 49, no. 1, pp. 1–16, Jan. 2001.

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[14] Q. Du, H. Ren, and C.-I. Chang, "A study between orthogonal subspace projection and generalized likelihood ratio test in hyperspectral image analysis," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jun. 2002, pp. 2575–2577.

[15] S. Kraut and L. L. Scharf, "The CFAR adaptive subspace detector is a scale-invariant GLRT," *IEEE Trans. Signal Process.*, vol. 47, no. 9, pp. 2538–2541, Sep. 1999.

[16] D. G. Manolakis, R. Lockwood, T. Cooley, and J. Jacobson, "Is there a best hyperspectral detection algorithm?" in *Proc. SPIE*, vol. 7334, Apr. 2009, Art. no. 733402.

[17] D. G. Manolakis, G. A. Shaw, and N. Keshava, "Comparative analysis of hyperspectral adaptive matched filter detectors," in *Proc. SPIE*, vol. 4049, Aug. 2000, pp. 2–17.

[18] L. Gao, B. Yang, Q. Du, and B. Zhang, "Adjusted spectral matched filter for target detection in hyperspectral imagery," *Remote Sens.*, vol. 7, no. 6, pp. 6611–6634, 2015.

[19] S. Kraut, L. L. Scharf, and R. W. Butler, "The adaptive coherence estimator: A uniformly most-powerful-invariant adaptive detection statistic," *IEEE Trans. Signal Process.*, vol. 53, no. 2, pp. 427–438, Feb. 2005.

[20] W. H. Farrand and J. C. Harsanyi, "Mapping the distribution of mine tailings in the Coeur d'Alene river valley, Idaho, through the use of a constrained energy minimization technique," *Remote Sens. Environ.*, vol. 59, no. 1, pp. 64–76, Jan. 1997.

[21] J. Theiler and B. R. Foy, "Effect of signal contamination in matched-filter detection of the signal on a cluttered background," *IEEE Geosci. Remote Sens. Lett.*, vol. 3, no. 1, pp. 98–102, Jan. 2006.

[22] W. F. Basener, "Clutter and anomaly removal for enhanced target detection," in *Proc. SPIE*, Apr. 2010, Art. no. 769525.

[23] J. C. Harsanyi and C.-I. Chang, "Hyperspectral image classification and dimensionality reduction: An orthogonal subspace projection approach," *IEEE Trans. Geosci. Remote Sens.*, vol. 32, no. 4, pp. 779–785, Jul. 1994.

[24] J. C. Harsanyi, "Detection and classification of subpixel spectral signatures in hyperspectral image sequences," Ph.D. dissertation, Dept. Elect. Eng., Univ. Maryland-Baltimore County, Baltimore, MD, USA, 1993.

[25] C. Wang *et al.*, "Unsupervised orthogonal subspace projection approach to magnetic resonance image classification," *Opt. Eng.*, vol. 41, no. 7, pp. 1546–1557, 2002.

[26] Y. Du, C.-I. Chang, H. Ren, C.-C. Chang, J. O. Jensen, and F. M. D'Amico, "New hyperspectral discrimination measure for spectral characterization," *Opt. Eng.*, vol. 43, no. 8, pp. 1777–1786, 2004.

[27] Y. Chen, N. M. Nasrabadi, and T. D. Tran, "Simultaneous joint sparsity model for target detection in hyperspectral imagery," *IEEE Geosci. Remote Sens. Lett.*, vol. 8, no. 4, pp. 676–680, Jul. 2011.

[28] Q. Du, H. Ren, and C.-I. Chang, "A comparative study for orthogonal subspace projection and constrained energy minimization," *IEEE Trans. Geosci. Remote Sens.*, vol. 41, no. 6, pp. 1525–1529, Jun. 2003.

[29] D. C. Heinz and C.-I. Chang, "Fully constrained least squares linear spectral mixture analysis method for material quantification in hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 39, no. 3, pp. 529–545, Mar. 2001.

[30] H. Kwon and N. M. Nasrabadi, "A comparative analysis of kernel subspace target detectors for hyperspectral imagery," *EURASIP J. Adv. Signal Process.*, vol. 2007, no. 1, pp. 1–13, Dec. 2006.

[31] L. Zhang, L. Zhang, D. Tao, and X. Huang, "Sparse transfer manifold embedding for hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 2, pp. 1030–1043, Feb. 2013.

[32] F. Verdoja and M. Grangetto, "Graph Laplacian for image anomaly detection," *Mach. Vis. Appl.*, vol. 31, nos. 1–2, pp. 1–6, Feb. 2018.

[33] D. Zhu, B. Du, and L. Zhang, "Target dictionary construction-based sparse representation hyperspectral target detection methods," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 4, pp. 1254–1264, Apr. 2019.

[34] H. Kwon and N. M. Nasrabadi, "Kernel matched subspace detectors for hyperspectral target detection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 28, no. 2, pp. 178–194, Feb. 2005.

[35] W. Li, Q. Du, and B. Zhang, "Combined sparse and collaborative representation for hyperspectral target detection," *Pattern Recognit.*, vol. 48, no. 12, pp. 3904–3916, 2015.

[36] Y. Liu, G. Gao, and Y. Gu, "Tensor matched subspace detector for hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 4, pp. 1967–1974, Apr. 2016.

[37] X. Yang, J. Chen, and Z. He, "Sparse-SpatialCEM for hyperspectral target detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 7, pp. 2184–2195, Jul. 2019.

[38] G. Zhang, S. Zhao, W. Li, Q. Du, Q. Ran, and R. Tao, "HTD-Net: A deep convolutional neural network for target detection in hyperspectral imagery," *Remote Sens.*, vol. 12, no. 9, p. 1489, 2020.

[39] Y. Shi, J. Li, Y. Zheng, B. Xi, and Y. Li, "Hyperspectral target detection with RoI feature transformation and multiscale spectral attention," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 6, pp. 5071–5084, Jun. 2020.

[40] T. Cheng and B. Wang, "Decomposition model with background dictionary learning for hyperspectral target detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 1872–1884, 2021.

[41] L. Zhang, L. Zhang, D. Tao, X. Huang, and B. Du, "Hyperspectral remote sensing image subpixel target detection based on supervised metric learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 8, pp. 4955–4965, Aug. 2013.

[42] A. Baterina and C. Oppus, "Image edge detection using ant colony optimization," *WSEAS Trans. Signal Process.*, vol. 6, no. 2, pp. 58–67, 2010.

[43] C. Ravazzi, G. Coluccia, and E. Magli, "Curl-constrained gradient estimation for image recovery from highly incomplete spectral data," *IEEE Trans. Image Process.*, vol. 26, no. 6, pp. 2656–2668, Jun. 2017.

[44] Y. Boykov and G. Funka-Lea, "Graph cuts and efficient N-D image segmentation," *Int. J. Comput. Vis.*, vol. 70, no. 2, pp. 109–131, Nov. 2006.

[45] J. Santner, T. Pock, and H. Bischof, "Interactive multi-label segmentation," in *Proc. Asian Conf. Comput. Vis.* Queenstown, New Zealand: Springer, 2010, pp. 397–410.

[46] A. Sandryhaila and J. Moura, "Discrete signal processing on graphs: Graph Fourier transform," in *Proc. IEEE Int. Conf. Acoust. Speech Signal Process.*, May 2013, pp. 6167–6170.

[47] D. I. Shuman, S. K. Narang, P. Frossard, A. Ortega, and P. Vandergheynst, "The emerging field of signal processing on graphs: Extending high-dimensional data analysis to networks and other irregular domains," *IEEE Signal Process. Mag.*, vol. 30, no. 3, pp. 83–98, May 2013.

[48] A. Sandryhaila and J. M. F. Moura, "Big data analysis with signal processing on graphs: Representation and processing of massive data sets with irregular structure," *IEEE Signal Process. Mag.*, vol. 31, no. 5, pp. 80–90, Sep. 2014.

[49] Y. Yuan, D. Ma, and Q. Wang, "Hyperspectral anomaly detection by graph pixel selection," *IEEE Trans. Cybern.*, vol. 46, no. 2, pp. 3123–3134, Dec. 2015.

[50] D. Messinger and J. Albano, "A graph theoretic approach to anomaly detection in hyperspectral imagery," in *Proc. 3rd Workshop Hyperspectral Image Signal Process. Evol. Remote Sens. (WHISPERS)*, 2011, pp. 1–4.

[51] W. Li and Q. Du, "Collaborative representation for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 3, pp. 1463–1474, Mar. 2015.

[52] L. Grady and J. R. Polimeni, *Discrete Calculus: Applied Analysis on Graphs for Computational Science*, vol. 3. New York, NY, USA: Springer-Verlag, 2010.

[53] C. Zhang and D. D Florêncio, "Analyzing the optimality of predictive transform coding using graph-based models," *IEEE Signal Process. Lett.*, vol. 20, no. 1, pp. 106–109, 2012.

[54] H. Kwon and N. M. Nasrabadi, "Kernel spectral matched filter for hyperspectral imagery," *Int. J. Comput. Vis.*, vol. 71, no. 2, pp. 127–141, Feb. 2007.

[55] X. Zhao, Z. Hou, X. Wu, W. Li, P. Ma, and R. Tao, "Hyperspectral target detection based on transform domain adaptive constrained energy minimization," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 103, Dec. 2021, Art. no. 102461.

[56] W. Li and Q. Du, "A survey on representation-based classification and detection in hyperspectral remote sensing imagery," *Pattern Recognit. Lett.*, vol. 83, no. 2, pp. 115–123, Nov. 2016.

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